

CLIMATE CHANGE ADAPTATION & RESILIENCE

Climate change resilience and adaptation of a population involve the ability to anticipate, adapt to, and recover from climate-related shocks through deliberate policies, procedures, and protocols. Even though climate change is a global issue, its effects are felt differently in different regions across the world. Two international situations of increasingly alarming extreme-weather events are the cases of urban heat islands in the Republic of Korea (ROK) and flooding and landslides in Rwanda. They are connected through the changes in the hydrological cycle that affect not only the two countries but also have lasting effects on other areas around the world.



Mission Statement & Methodology



To harness space applications for the development of adaptation and resilience strategies to climate change-induced challenges by analyzing case studies in the ROK and Rwanda. The aim is to provide interdisciplinary, inclusive, and sustainable recommendations to empower communities. Key stakeholders that are addressed are the local and national governments of the Republic of Korea and Rwanda, especially in their disaster management and environment sections. Roadmaps are provided to detail the possible timeline for the implementation of large-scale changes.

Information on the current state of climate change, its effect on the population and economy, disaster management, and existing policy frameworks in the investigated countries was gathered through an extensive literature review. Geographic Information System (GIS) data was amassed and mapped to visualize spatial and temporal differences of temperature, vegetation, population, and social vulnerability in Seoul between 2014 and 2024. By critically analyzing the gathered data, gaps were identified in the existing resilience strategies of the ROK and Rwanda. Both policy and technological proposals were provided to key stakeholders to enable improved disaster resilience using space-based systems.

Key findings

The unprecedented severity of Urban Heat Islands (UHI) in the ROK is driven by climate shifts, rapid urbanization, and land surface changes. These temperature anomalies heighten heat vulnerability by compounding public health risks, overburdening infrastructure, and deepening systemic inequalities. Specifically, UHIs in Seoul and other urban centers are the consequence of environmental and anthropogenic factors: low-albedo and heat-retaining urban materials, dense urban geometry, and reduced vegetation cover. Current monitoring systems in the ROK lack the spatial resolution needed for neighborhood-scale heat mapping and fail to integrate socio-demographic metrics that capture both heat vulnerability and social vulnerability.

In the case of Rwanda, key findings pertain to the dramatic differences in climate change effects between the regions of Rwanda. Disaster related indices are key tools in the quantification of at-risk areas. Low elevation and high population density areas have the highest disaster vulnerability. Natural disasters have cost over 1,000 deaths and over 25,000 destroyed homes over the last five years. Annually, the consequences of floods and landslides cost the country of Rwanda approximately 6% of its GDP. With 30% of Rwanda's GDP being supplied by the agricultural sector, which employs more than 60% of the country's population, flooding and landslide events cause an even greater disruption of the economy due to their destructive nature. The existing national policy framework of Rwanda provides a strong base for the country. Current satellite-based technologies employed by Rwanda are in the early stages of continuous use and the nation still lacks autonomous access to space-based data that is key in future disaster management.

Recommendations

This report recommends the use of an **Integrated Satellite-Based Resilience Framework**, which informs, monitors, and tracks heat hazards and vulnerabilities to mitigate UHI impacts in the ROK. The use of satellite remote sensing for **high-resolution optical and thermal infrared imagery** and detection of UHI intensity and Urban Thermal Field Variance Index (UTFVI) at neighborhood scale will be the development of the final proposed solution. **GNSS and ground sensors** will provide calibration and validation of satellite-derived land surface temperature (LST) products using precise location/timing, air temperature, humidity, and radiation data.

GIS and demographic/mobility data will integrate thermal and land cover datasets with sensitivity and adaptive capacity metrics to produce **satellite-based heat vulnerability maps**. **AI approaches** will be used for automated cloud-gap filling in LST datasets, real-time hotspot detection, diurnal heat risk mapping, and predictive modeling of seasonal and event-based UHI patterns.

The **mapping of multiple disaster indicators**, such as INFORM and the Flood Resilience Index, using satellite technology will be a key tool in policy and decision-making for resilience all over Rwanda, aiding in prioritizing the deployment of resources to high-risk areas. **Educating the general public** on climate change and disaster management can lead to increased openness to emergency protocols mandated by the government and improve the disaster response. Community workshops that both teach the local population and gather feedback on the **needs and already existing solutions used by local communities** will further help with prioritization of necessary resources in a disaster scenario. Rwanda can expand its space capabilities through the **acquisition of cost-effective earth observation data and leveraging global partnerships** for secondary payload opportunities. Technology transfer between developed countries and Rwanda can be enhanced by building upon existing schemes. A **novel technological solution** is presented that includes the deployment of landslide detection ground sensors that are connected over satellite communication and act as a local early warning system.

Conclusion

This report delves into how satellite-based solutions can be used to improve the climate change resilience of regions affected by an increasing number of disaster events. The two case studies were selected not only due to the severe challenges they face, but also because they reflect broader global climate dynamics. The ROK already employs solid resilience strategies such as the use of early warning systems. However, a focus should be placed on the improvement of EO spatial resolution to identify temperature hotspots in Seoul with higher precision and combine physical with demographic data to evaluate vulnerability hotspots around the city.

While Rwanda has a strong policy foundation, a lack of technical knowledge and international partnerships hinders the country from reaching its full potential concerning disaster management sooner. The country's progress in climate resilience can be accelerated through the technology transfer of space technology, increased education of the general public, and the expanded use of GIS data to efficiently inform policy and decision makers. Key stakeholders include the governments of the ROK and Rwanda, focusing specifically on their departments directed at the environment and disaster management, and international partners that support space-based ventures with the two countries.

